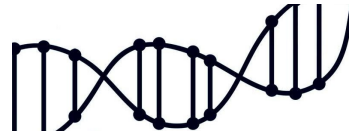




Introduction to **Evolutionary Computation and Deep Learning**



CJ Chung
Computer Science
Lawrence Technological University



EC-DL

Natural way to use bio-inspired algorithms to optimize neural nets & Deep Learning model architectures

What for?

Solving
challenge
problems in DL.
Publishing
Research
Papers

How?

Simulating
evolution of
neural
net models

Main Topics

1. Intro to Evolutionary Deep Learning (EDL)
2. Review of NN and DL: Perceptron, Deep Feedforward, CNN, Regression, Time-series models, **Multivariate output time series**)
3. Intro to Evolutionary Computation (EC) and optimization algorithms: GA, ES, ES with 1/5 rule, **CMA-ES**
4. Evolutionary computation with DEAP library
5. Neural Network connection weight training by evolutionary algorithms (instead of BP)
6. Evolutionary hyperparameter/architecture optimization (using Keras and “5.”)
7. Topics
 - Evolving HP/Architecture of **LLMs/Transformers** (lighter, faster, more adaptable LLMs)
 - Evolution of **Augmenting Topologies**
 - **Reinforcement Learning (RL)**, Deep RL, and EC
 - Evolutionary Meta, Few-Shot, Continual learning
 - Evolutionary Discovery of Lightweight models: **Evolutionary Quantization, Evolving BitNets**
 - **Evolutionary Feature Engineering**
 - **Agentic AI and EC**

Syllabus in PDF on Canvas

LTU 1660-202610_MCS5993_M

Fall 2025

Account

Dashboard

Courses

Calendar

Inbox

History

eHelp

Home

Announcements

Syllabus

Discussions

Grades

People

Zoom

Media Gallery

My Media

Getting Started

LTU Resources

Technical Support

Evolutionary Computation and Deep Learning

Announcements

Syllabus

Course Materials

Assignments

Grades

Grading

- 6 homework assignments (12%) – 2 percent for each
- 3 written closed book quizzes (33%) – Questions are mainly from homework assignments
- Final written closed book exam, comprehensive (30%)
- Term Project (25%)

Important Dates (See also Grading Section)

Sep 1	Labor Day. No class
Sunday, Sep 7	Last day to drop courses with refund. HW1 due by Sep 3, 7pm.
Sep 22, 24	Async. online class; No in-person class; No in-person office hours
Friday, Nov 21	Last day for "W" – Friday
Nov 26 - Nov 30	Thanksgiving break
Dec 15-18 Wed., Dec 17	Final Exam Week; Final Exam, 20:00-21:50, LTU Final Exam Schedule

Quiz date may not be announced.

Required Markdown Text Cell for Each File

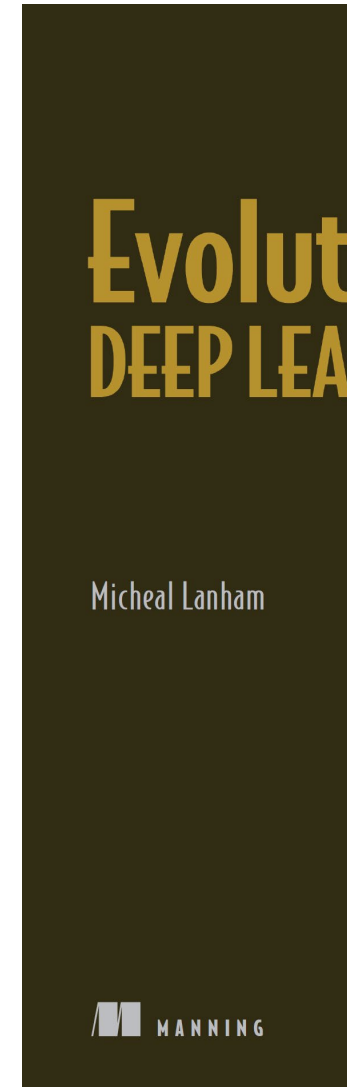
HW *n* Assignment Disclaimer

- **Author Name:** First Last
- **LTU ID:** 000000123
- **Work Ownership:** This work is my own. It is not copied from classmates (Yes/No):
- **AI Assistance:** Percentage of code generated with AI tools: xx%
(If greater than 0%, briefly describe how AI was used)
- **Understanding:** I understand every part of this code (Yes/No):
- **Confidence:** I am confident that I can modify, adapt, and extend this code on my own (Yes/No):

Recommended Textbook 1

Evolutionary Deep Learning:
Genetic algorithms and neural
networks

by [Micheal Lanham](#)



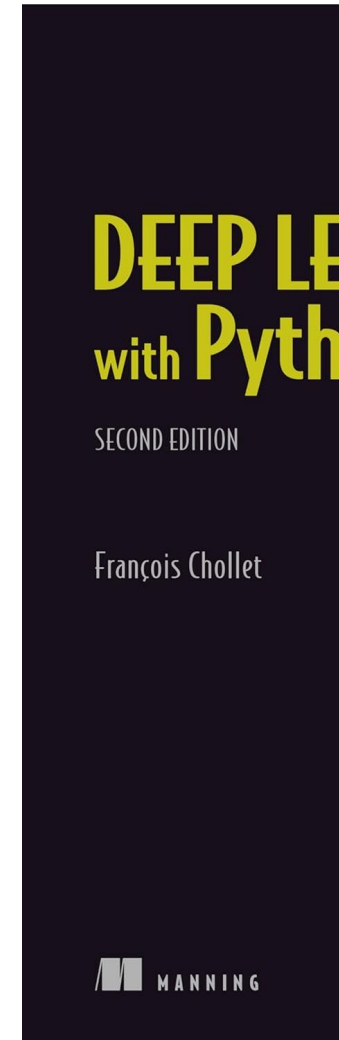
Genetic algorithms and neural networks



Recommended Textbook 2

- Author: François Chollet
- Deep Learning with Python, 2nd edition

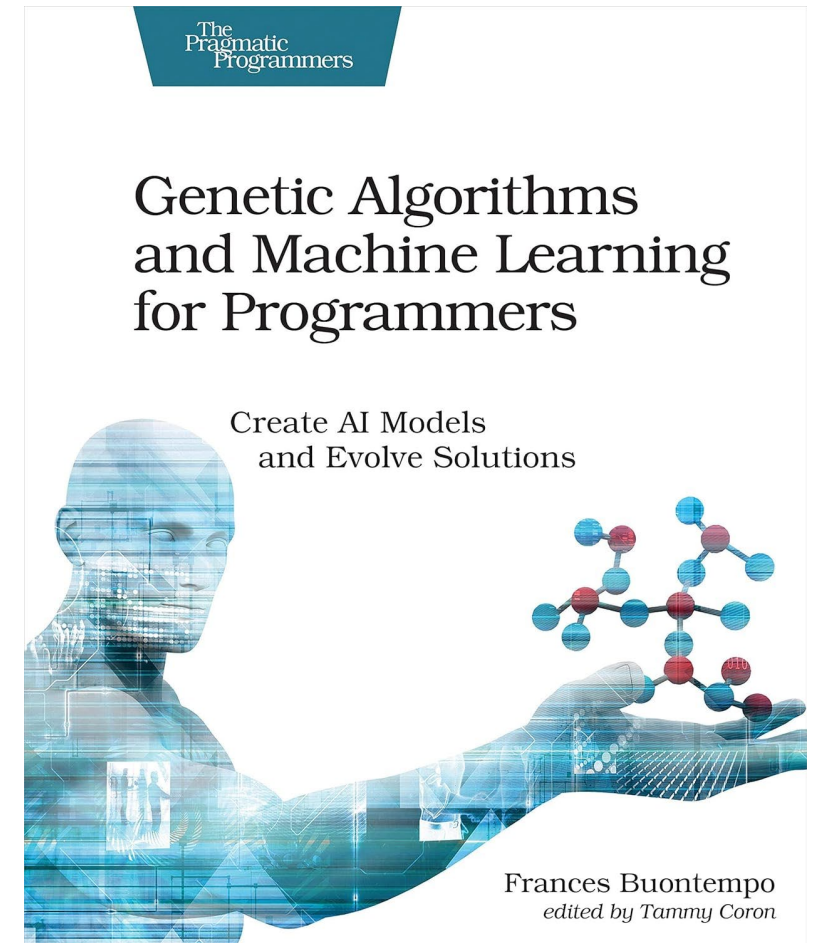
<https://www.manning.com/books/deep-learning-with-python-second-edition>



Recommended Textbook 3

Genetic Algorithms and Machine Learning for Programmers: Create AI Models and Evolve Solutions (Pragmatic Programmers) 1st Ed. by Frances Buontempo

Code in C++ ($\geq$ C++11), Python (2.x or 3.x) and JavaScript (using the HTML5 canvas)



Syllabus

- Any questions?
- Is everything clear?

INTRODUCTION TO AI

The Birthplace of AI at Dartmouth College, Hanover, New Hampshire, in 1956

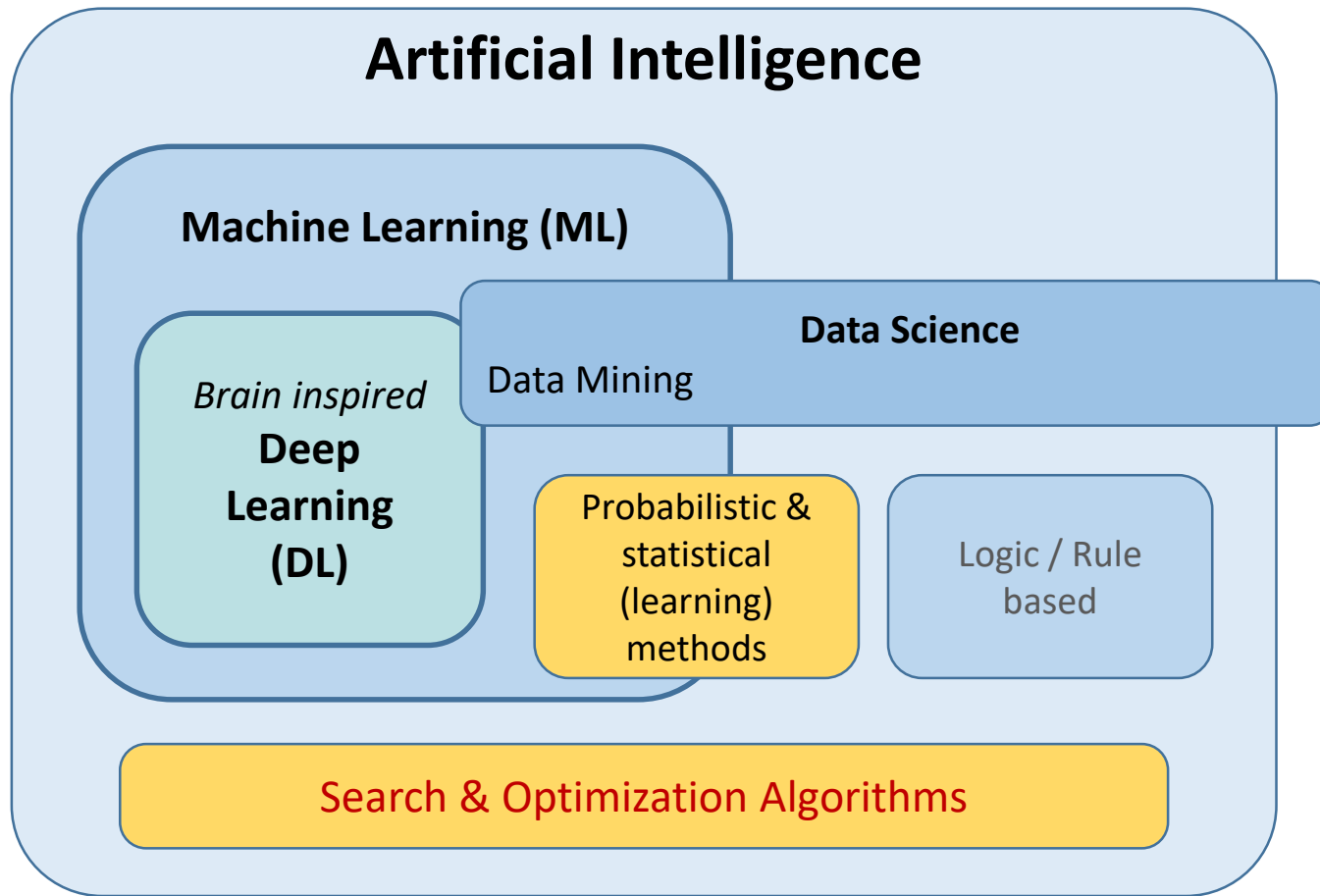


Seymour Papert



AI subfields

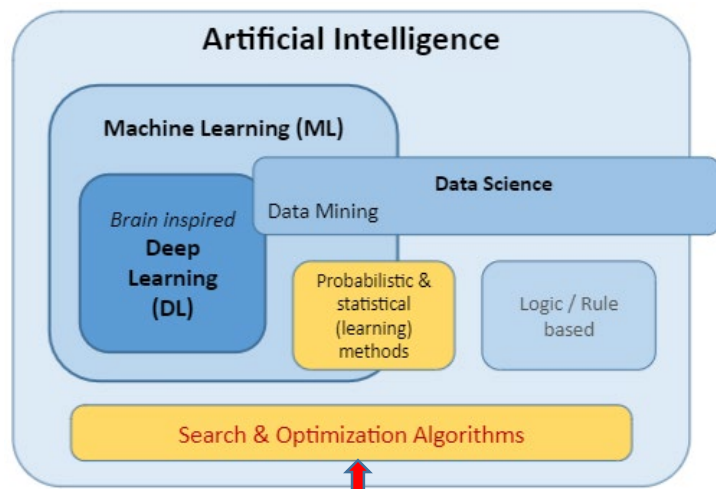
AI is to mimic human tasks (such as learning, adapting, and reacting) that require certain cognitive skills.



Data science is an interdisciplinary field focused on extracting knowledge from (large) data sets, and applying the knowledge to solve problems in a wide range of application domains.

Data mining is a process of extracting and discovering hidden patterns (rules) in large data sets

Field of Computer Science



Evolutionary Computation (EC)
Genetic Algorithms



1956



John Holland – Genetic Algorithms, 1975, U of M



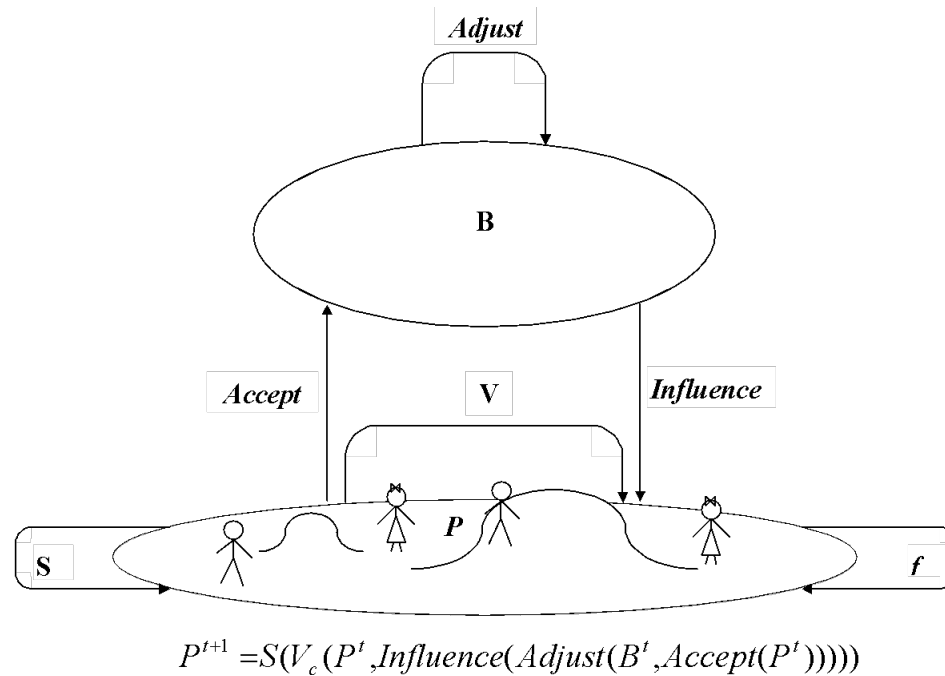
Robert Reynolds – Cultural Algorithms, 1994



CJ Chung – A Cultural Algorithm Framework, 1996

Chung's Cultural Algorithm Framework (1996)

A framework for optimization & adaptation with populations of
problem solvers and a belief space



Begin

$t = 0;$

Initialize P^t

Initialize B^t

repeat

Evaluate P^t

$Adjust(B^t, Accept(P^t))$

$Vary(P^t, Influence(B^t))$

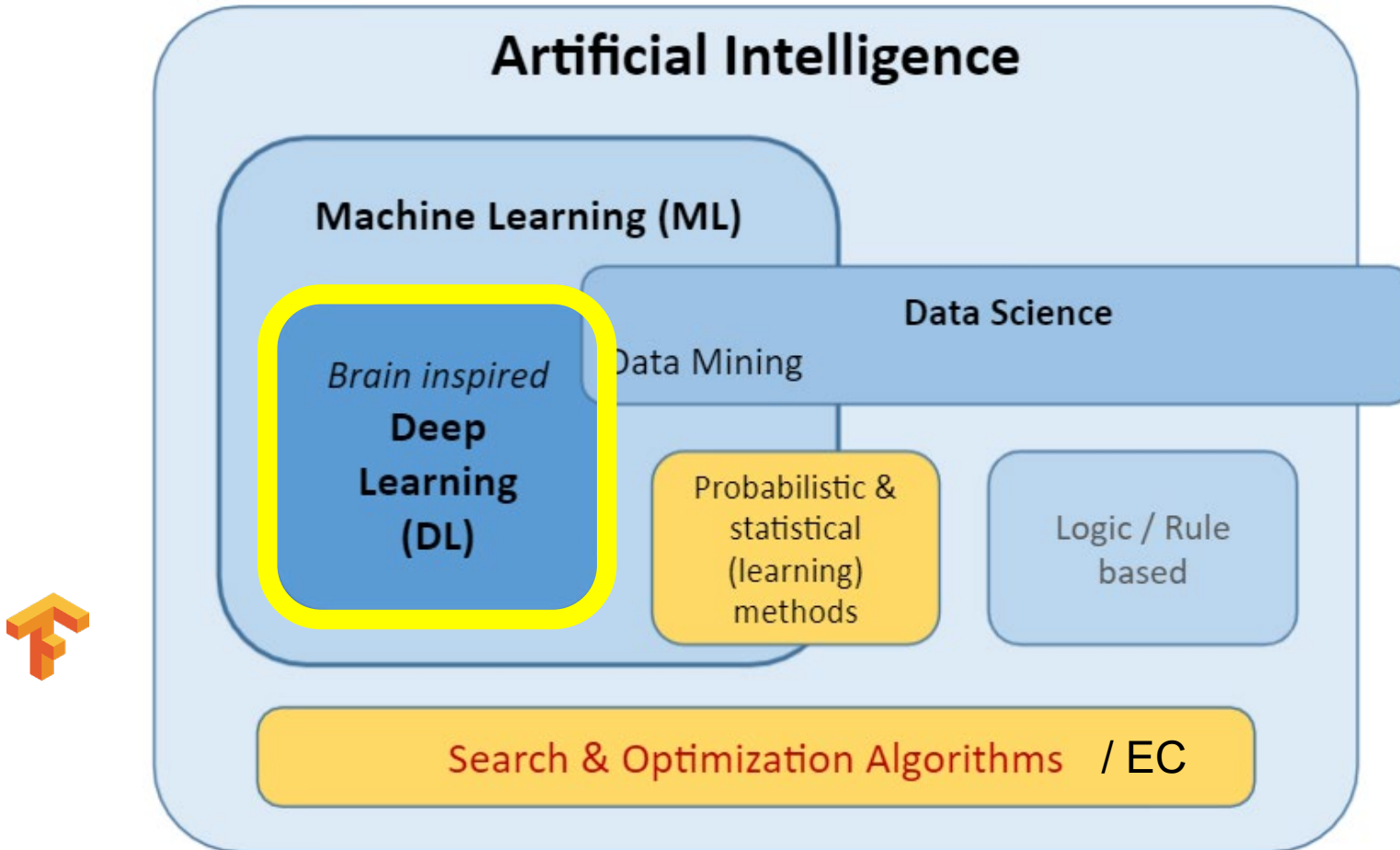
$t = t + 1;$

Select P^t from P^{t-1}

until (termination condition achieved)

End

AI popularity - CNN & Google's TensorFlow, a free ML/DL library in 2015

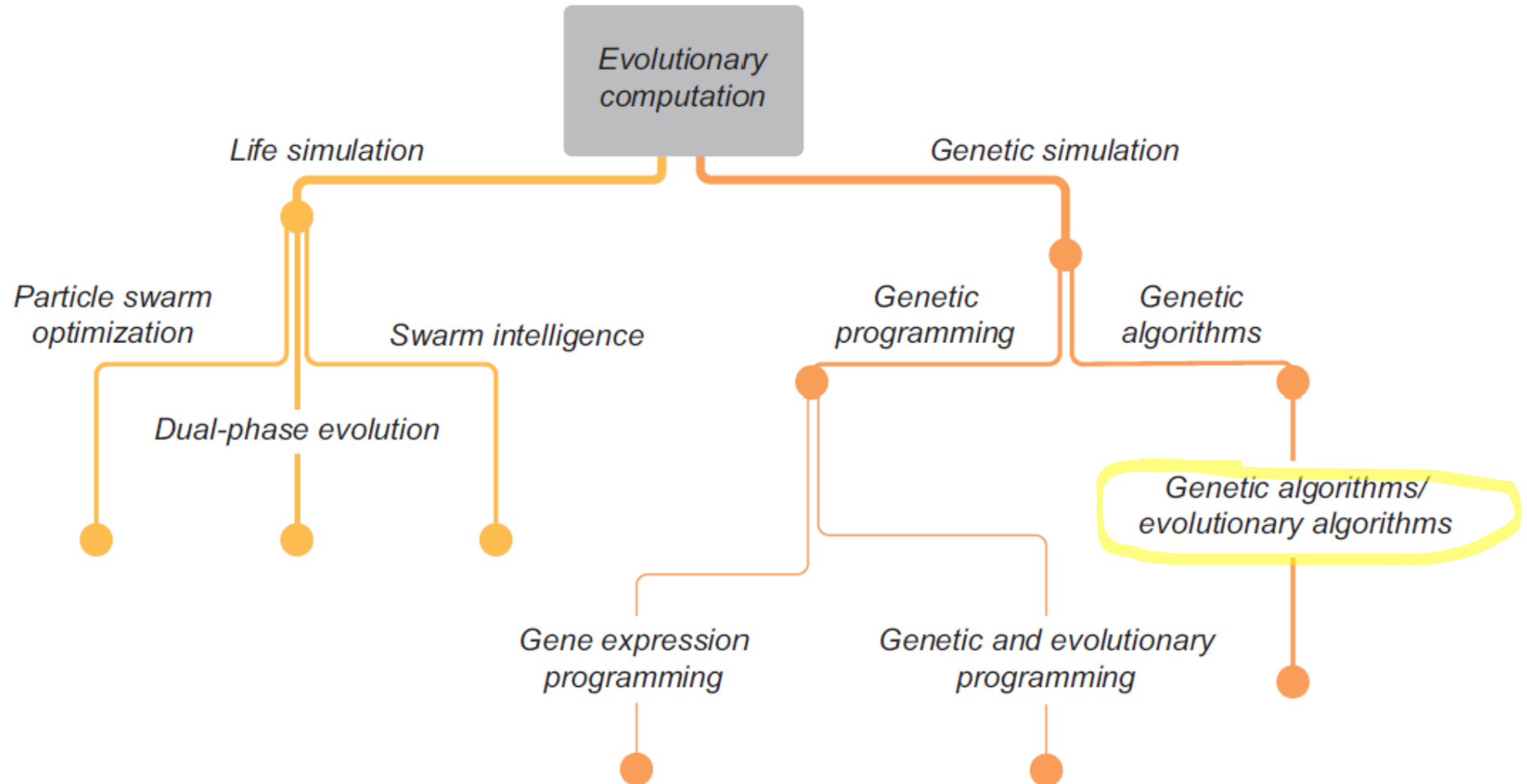


Brain-inspired DL is the most popular/powerful AI technology currently.

CNN: Convolutional neural network

INTRODUCTION TO EC

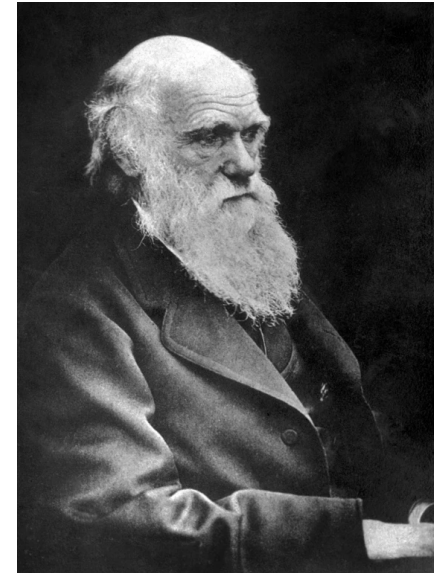
Evolutionary Computation, a subfield of AI

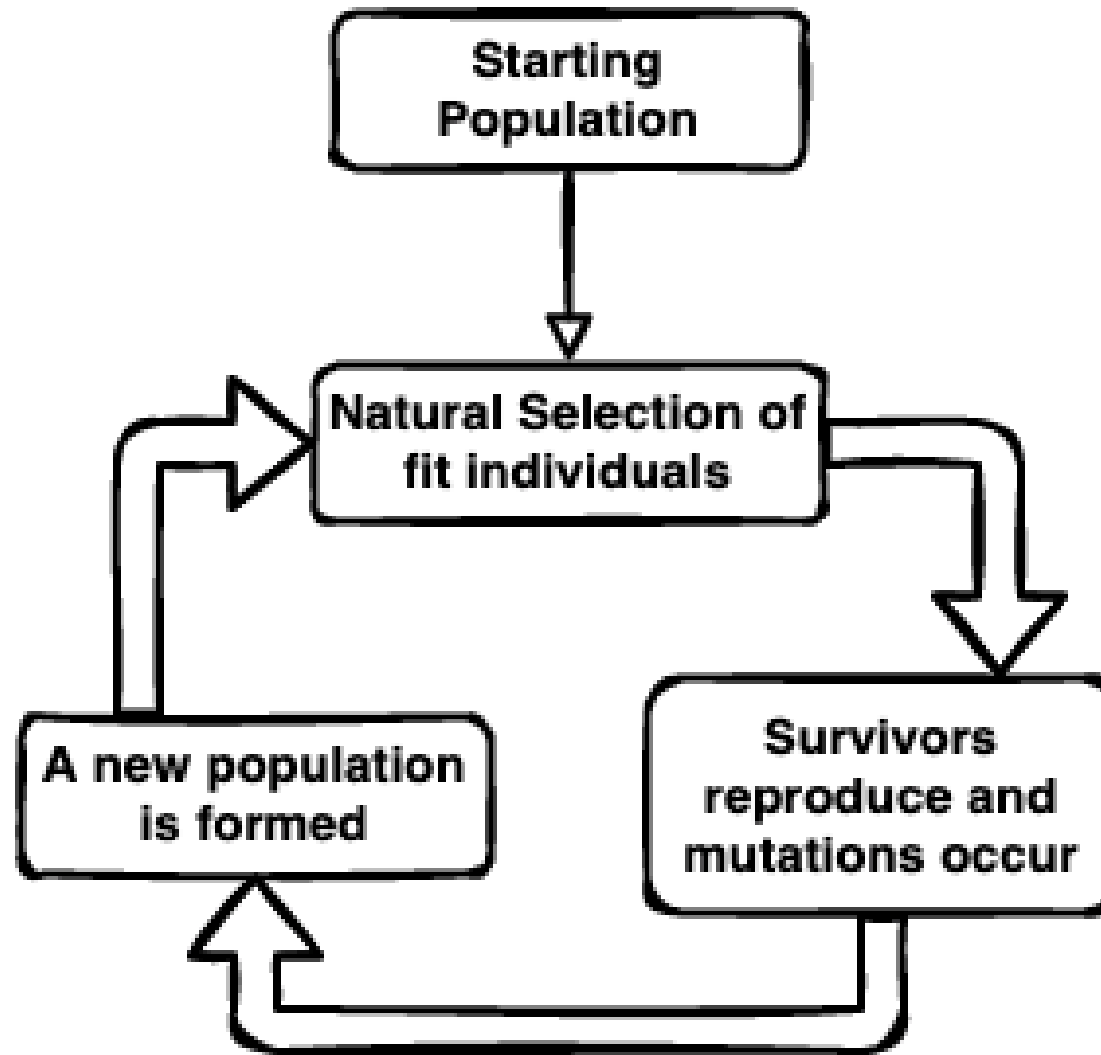


EA: *Learning* by Simulating Evolutions

The Origin of Species, Charles Darwin, 1859:
Survival of the Fittest

- Each individual tends to pass its traits to its offspring
- Nevertheless, nature produces individuals with different traits
- Survival of the fittest, driving the population toward favorable traits
- Over long periods, producing entirely new species whose traits make them especially suited to particular ecological niches





Genetic Algorithm (Evolutionary Algorithm)

initialize population

evaluate population

while TerminationCriteria_notSatisfied:

 select parents for reproduction

 perform recombination (xover) and mutation

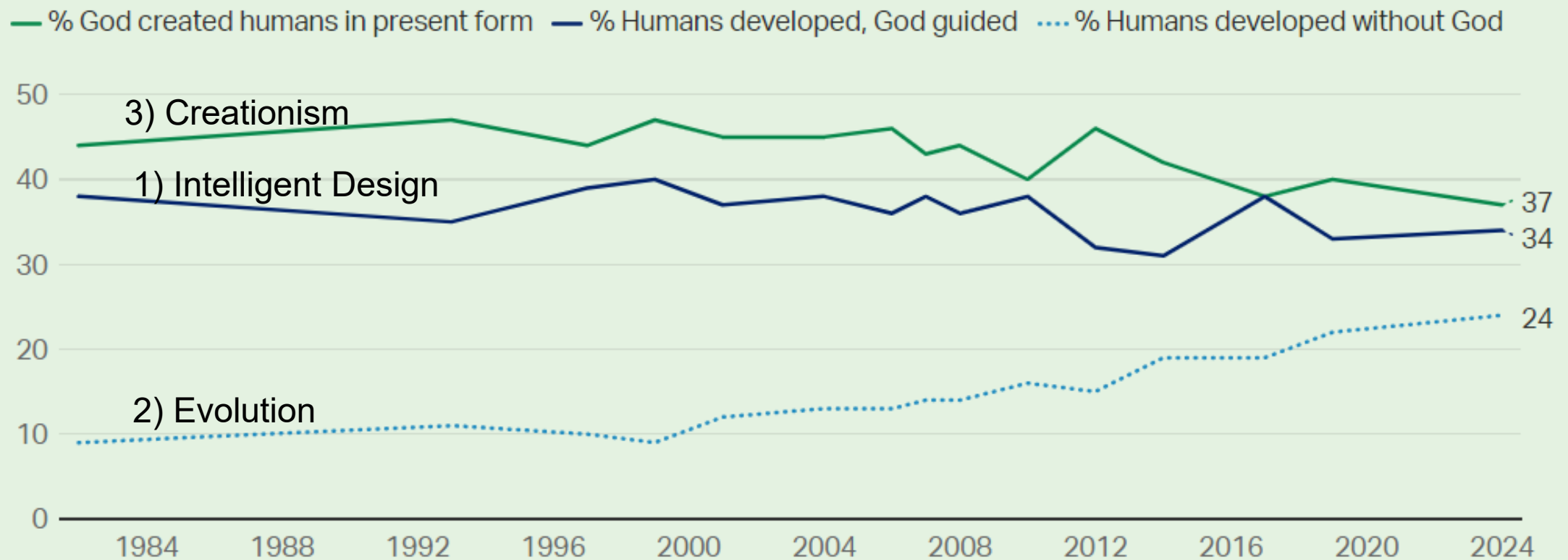
 evaluate population

Evolutionary algorithms: Guided random search algorithms for optimization problems

*By the way, do you believe in the theory of
evolution?*

Americans' Beliefs About Human Origins, 1982-2024

Which of the following statements comes closest to your views on the origin and development of human beings -- 1) Human beings have developed over millions of years from less advanced forms of life, but God guided this process, 2) Human beings have developed over millions of years from less advanced forms of life, but God had no part in this process, 3) God created human beings pretty much in their present form at one time within the last 10,000 years or so?



Anonymous Survey

<https://forms.gle/3MGt8qF6283gXnMW6>

- Human beings have developed over millions of years from less advanced forms of life, but God guided this process
- Human beings have developed over millions of years from less advanced forms of life, but God had no part in this process
- God created human beings pretty much in their present form at one time within the last 10,000 years or so



History of EA and EC

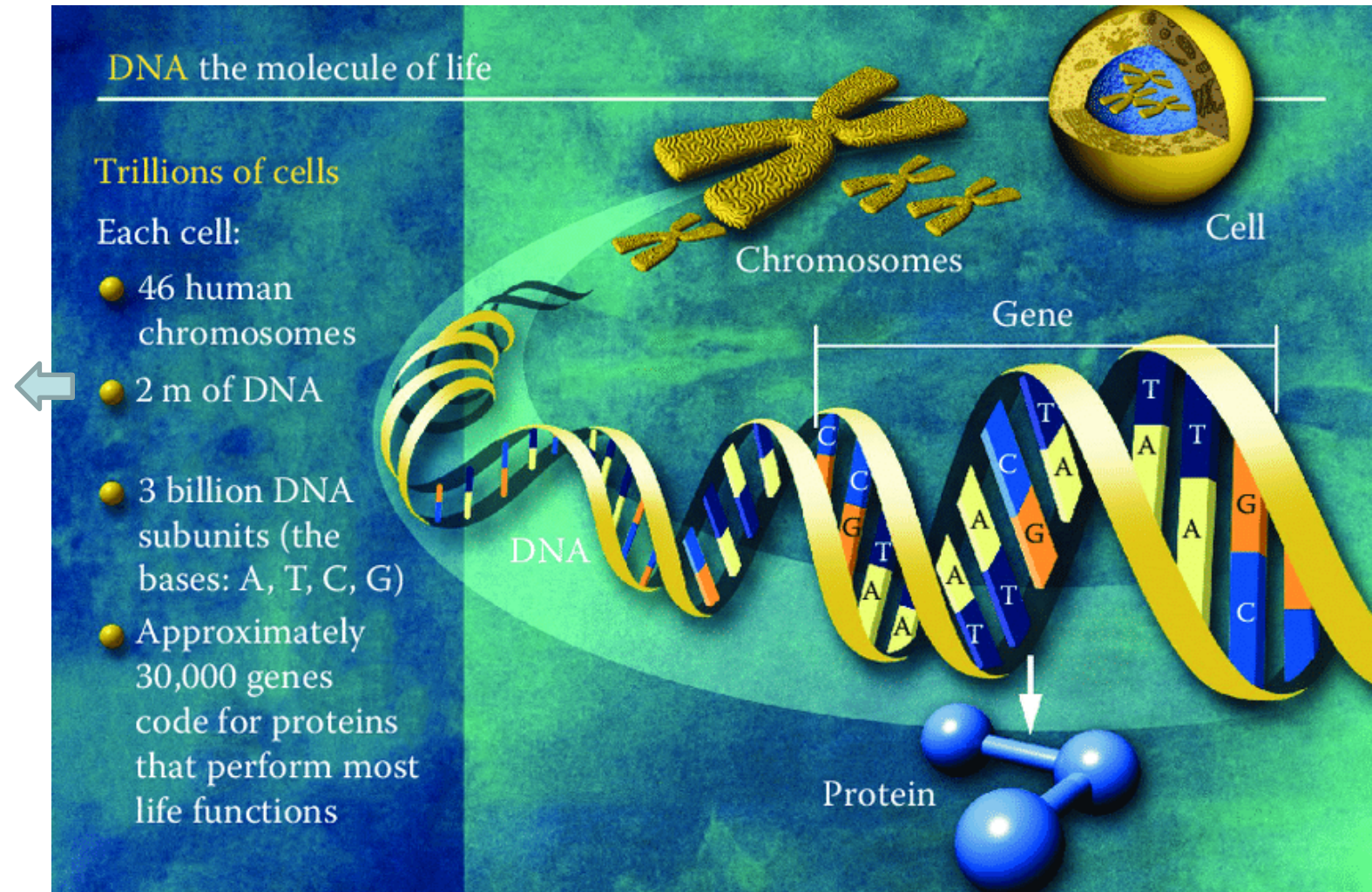
- Evolutionary Operation: Box (1951, 1957) – Experimental gradient method
- Genetic Algorithms: John Holland at U of M (1975) - Used binary representation
- Evolution Strategies: Rechenberg (1964, 1973)
- Evolutionary Programming, Fogel (1966)
- Genetic Programming, Koza (1992)
- ...
- Cultural Algorithms, Reynolds and Chung (1996)
- ...

Structure of DNA => GA

Each cell is packaged into 46 separate chromosomes (23 pairs)

Refers to the total length of DNA from all 46 chromosomes combined if they were fully unwound and stretched out end to end.

the DNA sequences in all the cells of a person are **nearly** identical.



Basically quaternary system, not ternary

Overview of the Genetic Algorithms

from a molecular point of view

- Chromosomes: the custodians of “trait determining factors” called genes
- Genes
- Operators
 - Reproduction: clone
 - Crossover: assembles existing genes into new combination
 - Mutation: produces new genes unseen so far
- Individual: chromosome (a candidate solution)
- Population: many individuals (candidate solutions)

Representation - Chromosomes could be:

- Binary Bit strings (0101 ... 1100)
- Integer numbers (1 2 4 6 ... 7 -2 5 9)
- Real numbers (43.0 -33.1 ... 0.0 89.2)
- Permutations of element (E11 E3 E7 ... E1 E15)
- Lists of rules (R1 R2 R3 ... R22 R23)
- Program elements (program1 program2 ...)
- Any data structure
 - Decision Trees
 - Neural Networks, CNNs, RNNs, ...
 - Graph NNs
 - LLMs / LFM
 - Robot Control software
 - ...

EDL

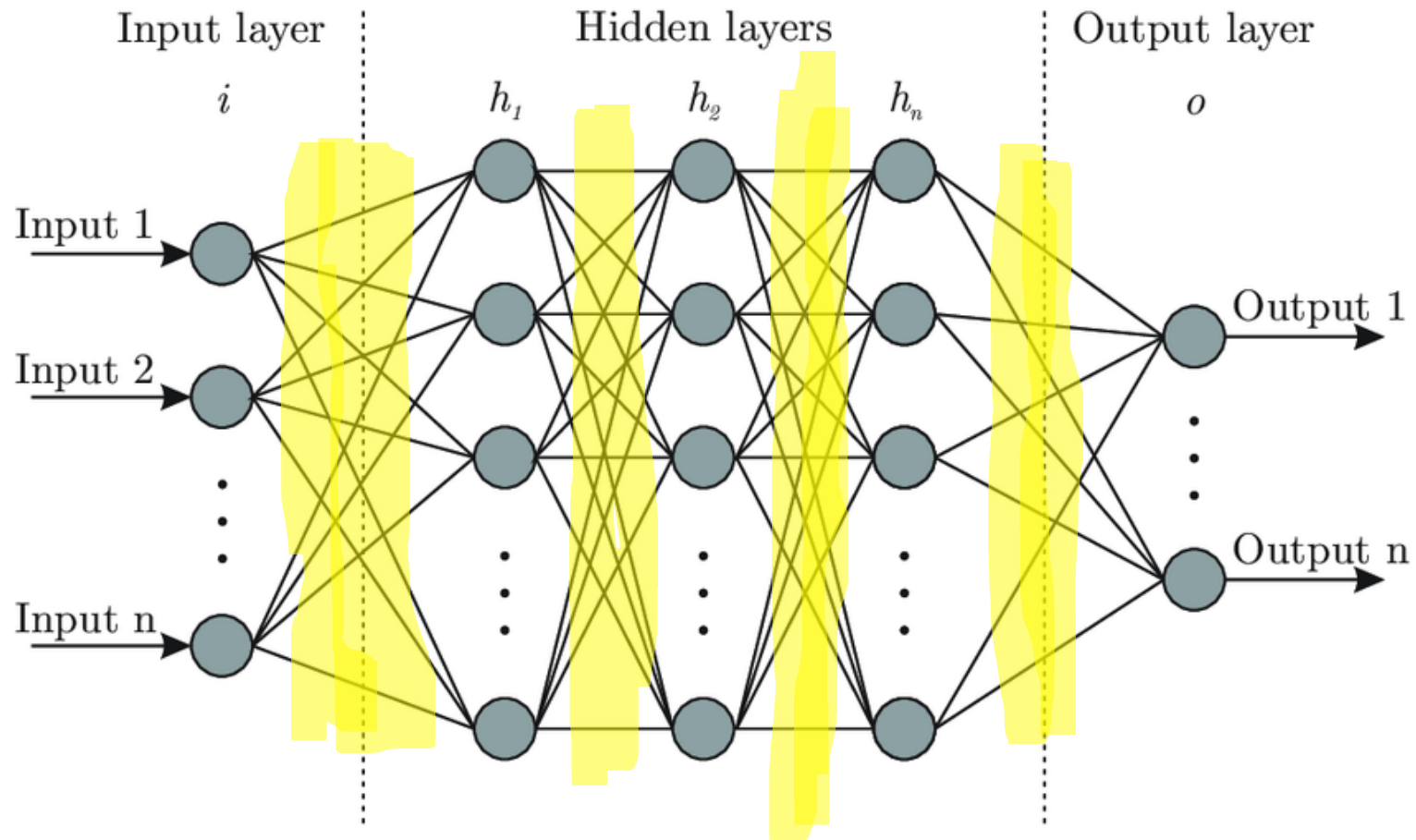
EDL (Evolutionary Deep Learning)

EDL is a broad term encompassing the combination of evolutionary methods with DL. Examples include:

1. Evolutionary Connection Weight (parameter) Optimization
(instead of Backpropagation)
2. Evolutionary Hyper-parameter / Architecture Optimization
Structure of neural networks
3. Neuro Evolution of Augmenting Topologies (NEAT)
Evolving both weights and architecture (1. + 2.). Also topologies
4. AutoML – to automate ML/DL model dev. workflow

Connection Weight (*Parameter*) Optimization

Optimizing weights (parameters) of a NN instead of using BP only



Evolutionary HyperParameter Optimization

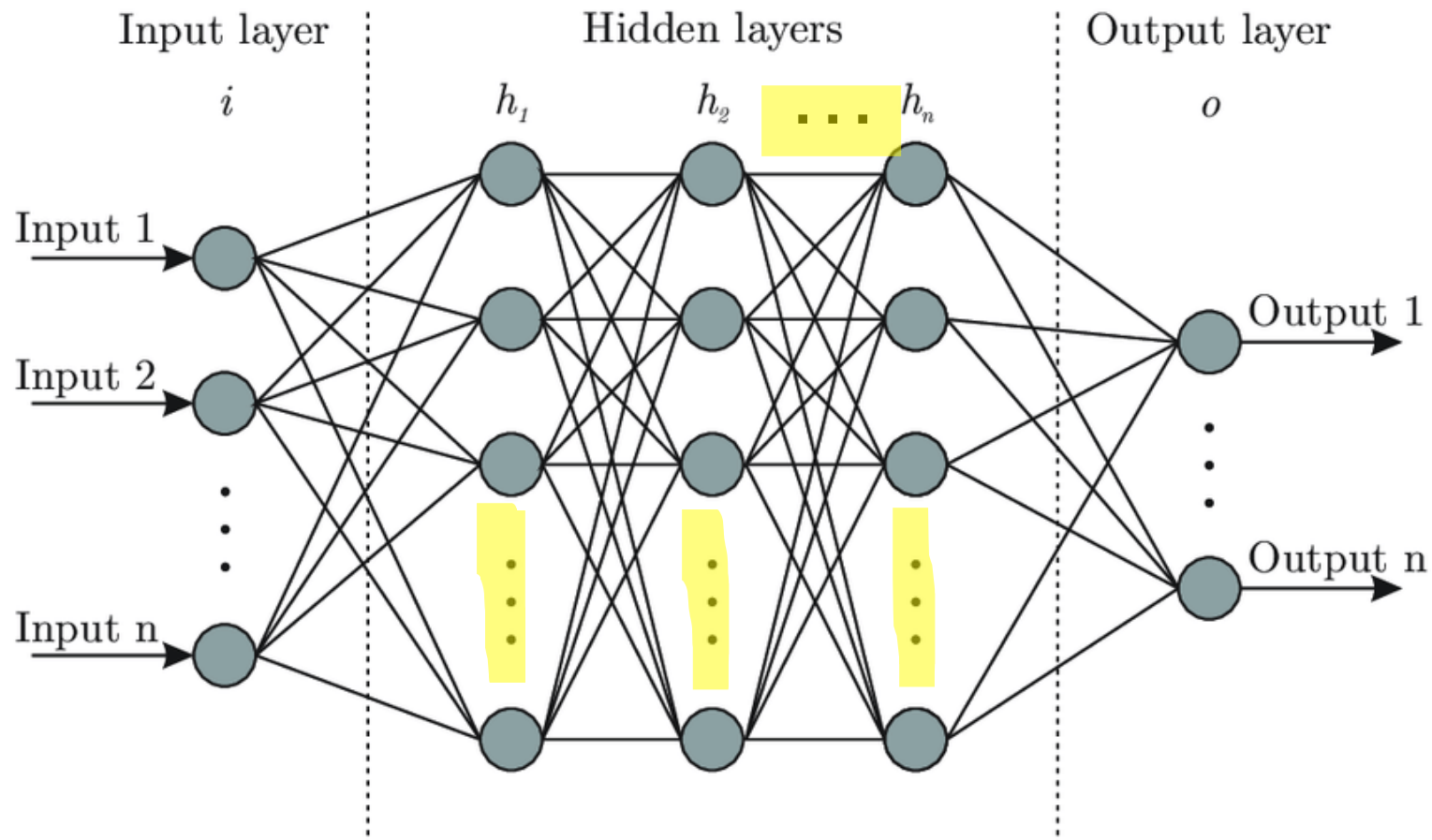
- *Hyperparameter tuning* is a process we undertake in ML/DL that requires the optimization of a model by tweaking the various control variables that define it.
- HPs include Learning rate, number of epochs, optimization method, loss method, dropout and normalization rates, size of layers, number of neurons for each layer, ...



Evolutionary Architecture Optimization (Next slide)

Evolutionary Architecture Optimization

Optimizing NN architecture, e.g. (1) number of hidden layers (2) number of neurons for each hidden layer



HPO Methods

- Grid Search: Define a discrete set of values for each hyperparameter. Try all combinations.
- Random Search: Sample hyperparameters randomly within predefined ranges. Often outperforms grid search for the same compute budget.
- Bayesian Optimization: Use a probabilistic model to select HPs based on acquisition function
- Population-based Guided Random Search → EC

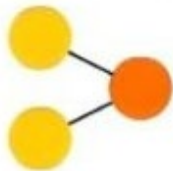
NEAT: Neuro Evolution of Augmenting Topologies

- NEAT is a technique that evolves both the weighting parameters and structures of networks.
- The NEAT approach begins with a perceptron-like feed-forward network of only input neurons and output neurons.
- As evolution progresses through discrete steps, the complexity of the network's topology may grow, either by inserting a new neuron into a connection path, or by creating a new connection between (formerly unconnected) neurons.

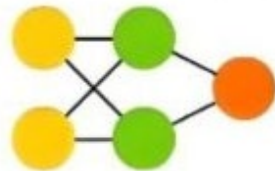
Topology in Neural Networks

- the architectural “map” of nodes and connections that defines how data moves through the model
- How the neurons are arranged and how they are connected to each other

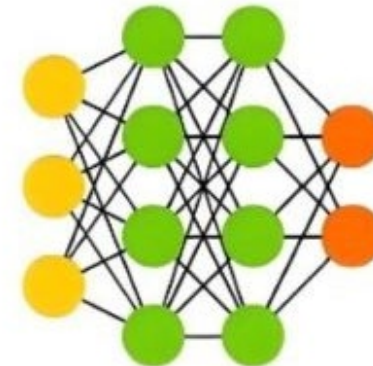
Perceptron (P)



Feed Forward (FF)

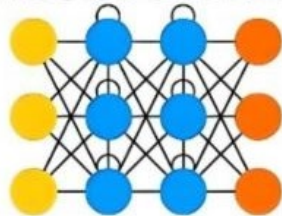


Deep Feed Forward (DFF)

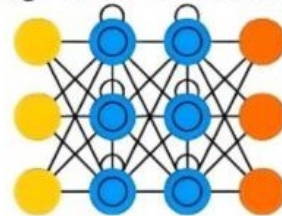




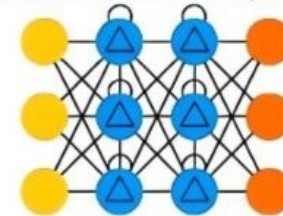
Recurrent Neural Network (RNN)



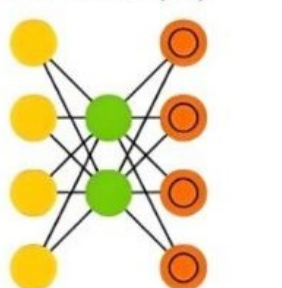
Long / Short Term Memory (LSTM)



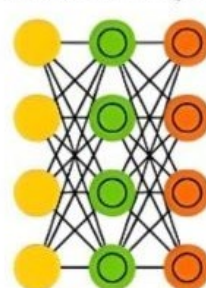
Gated Recurrent Unit (GRU)



Auto Encoder (AE)



Variational AE (VAE)



Denoising AE (DAE)



Sparse AE (SAE)



Markov Chain (MC)



Hopfield Network (HN)



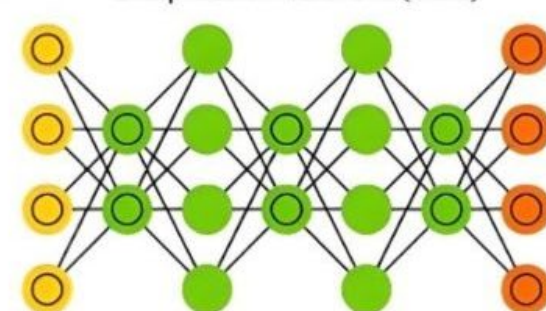
Boltzmann Machine (BM)

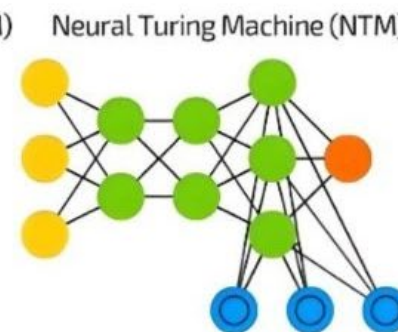
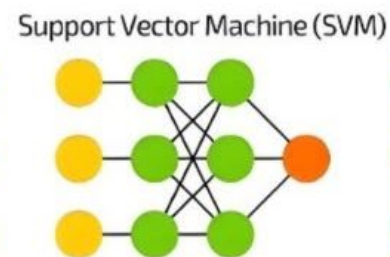
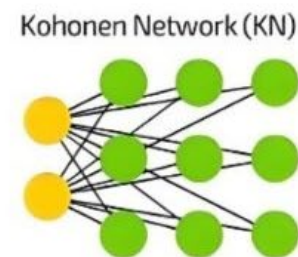
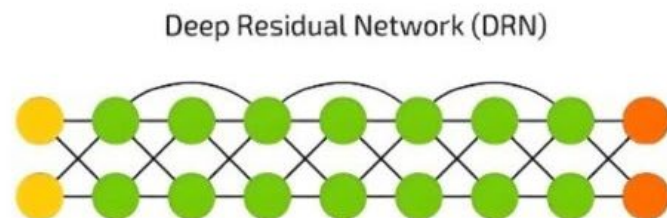
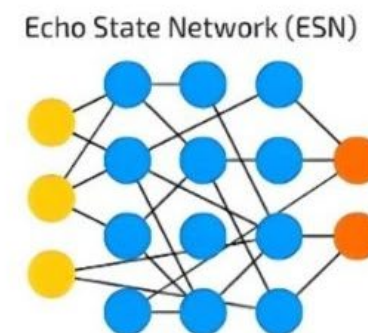
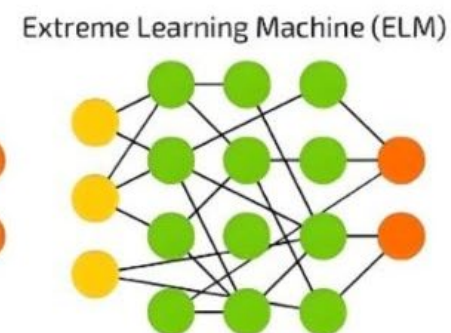
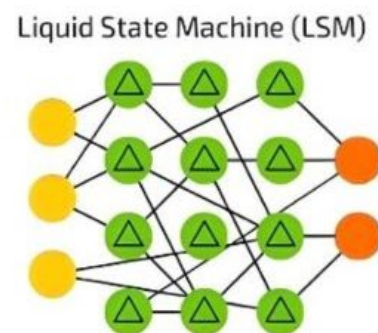
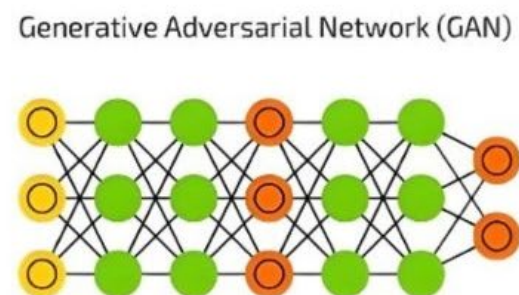
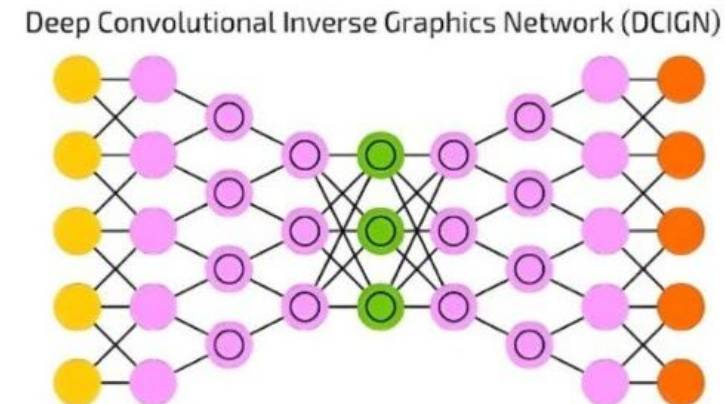
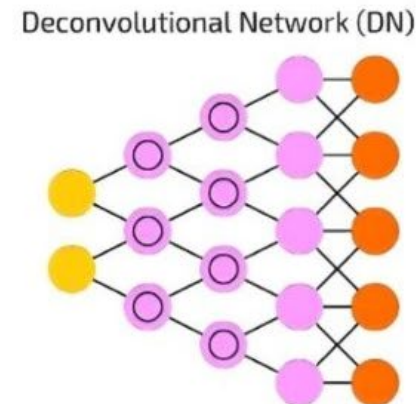
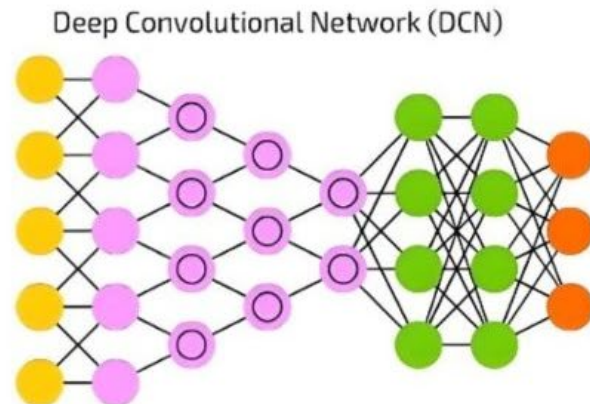


Restricted BM (RBM)



Deep Belief Network (DBN)

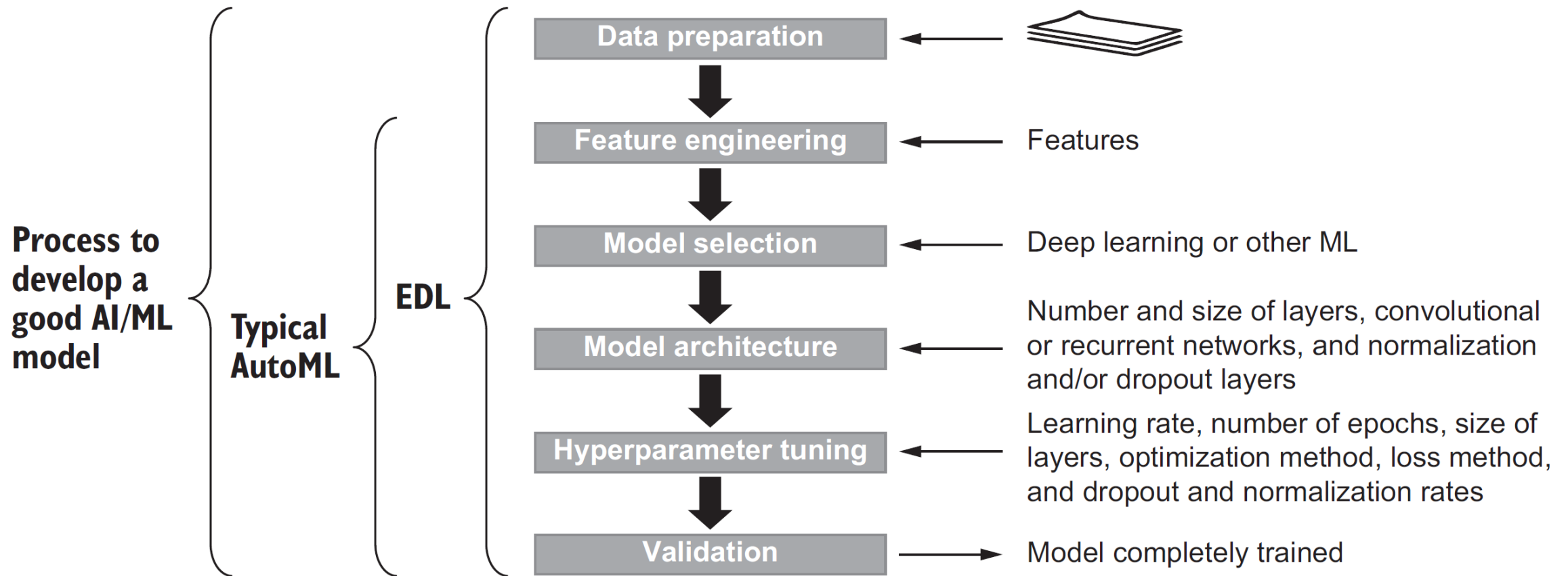




Auto ML

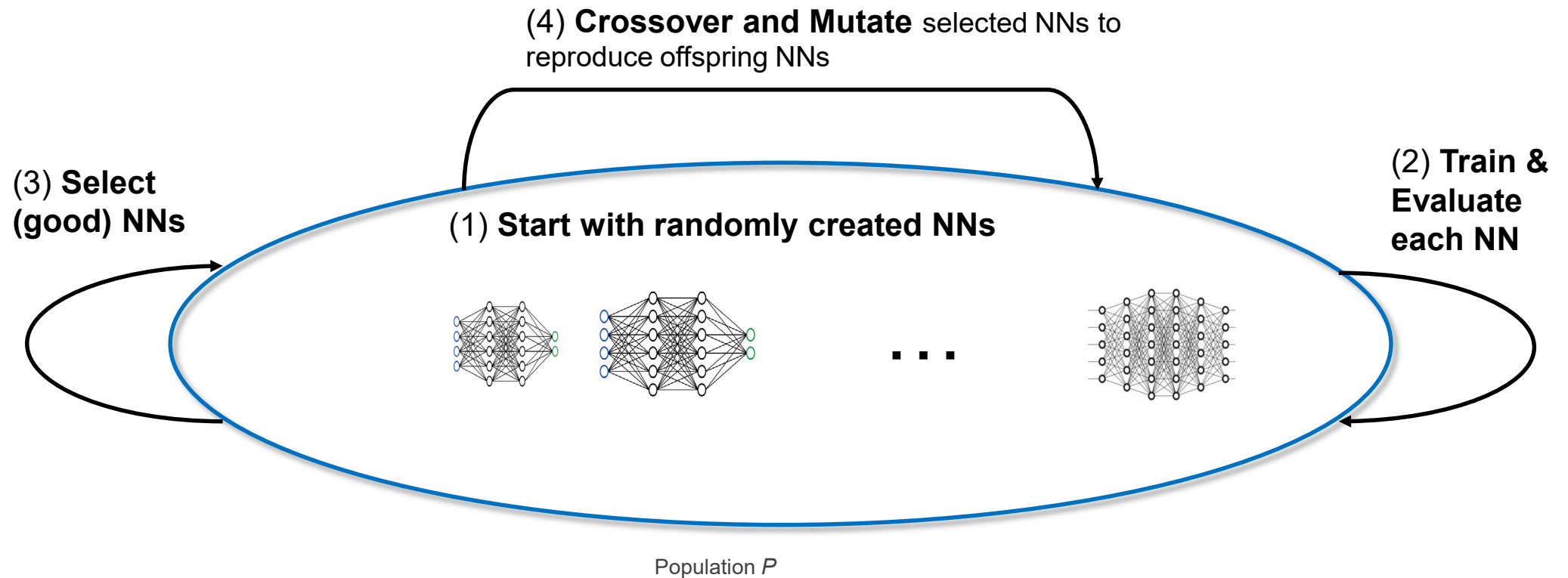
- Auto ML define a set of tools and techniques that look to automate the entire ML/DL model development workflow.
- Many forms of EC have been and can be used to automate the model development workflow.
- Google and other companies have invested significantly into developing Auto ML to assist consumers in building robust models for their own needs.
- While these services are powerful, they often work like a black box and limit more agile customization of new cutting-edge models.

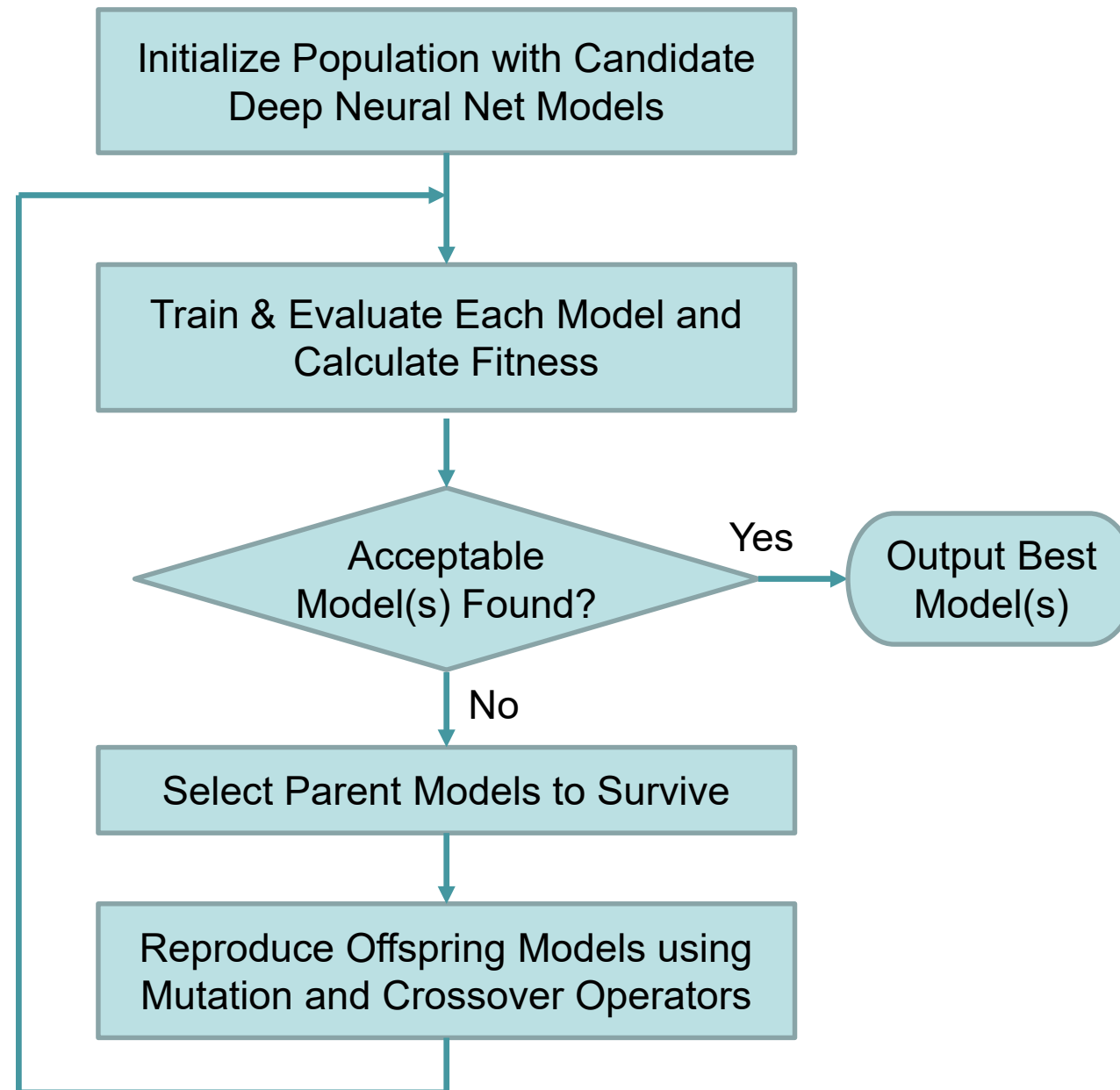
Steps for developing a good AI/ML model with AutoML and/or EDL



Evolutionary Deep Learning (EDL) is a hybrid approach that combines principles from evolutionary computation (EC) and deep learning (DL) to optimize neural network architectures, hyperparameters, and/or weights.

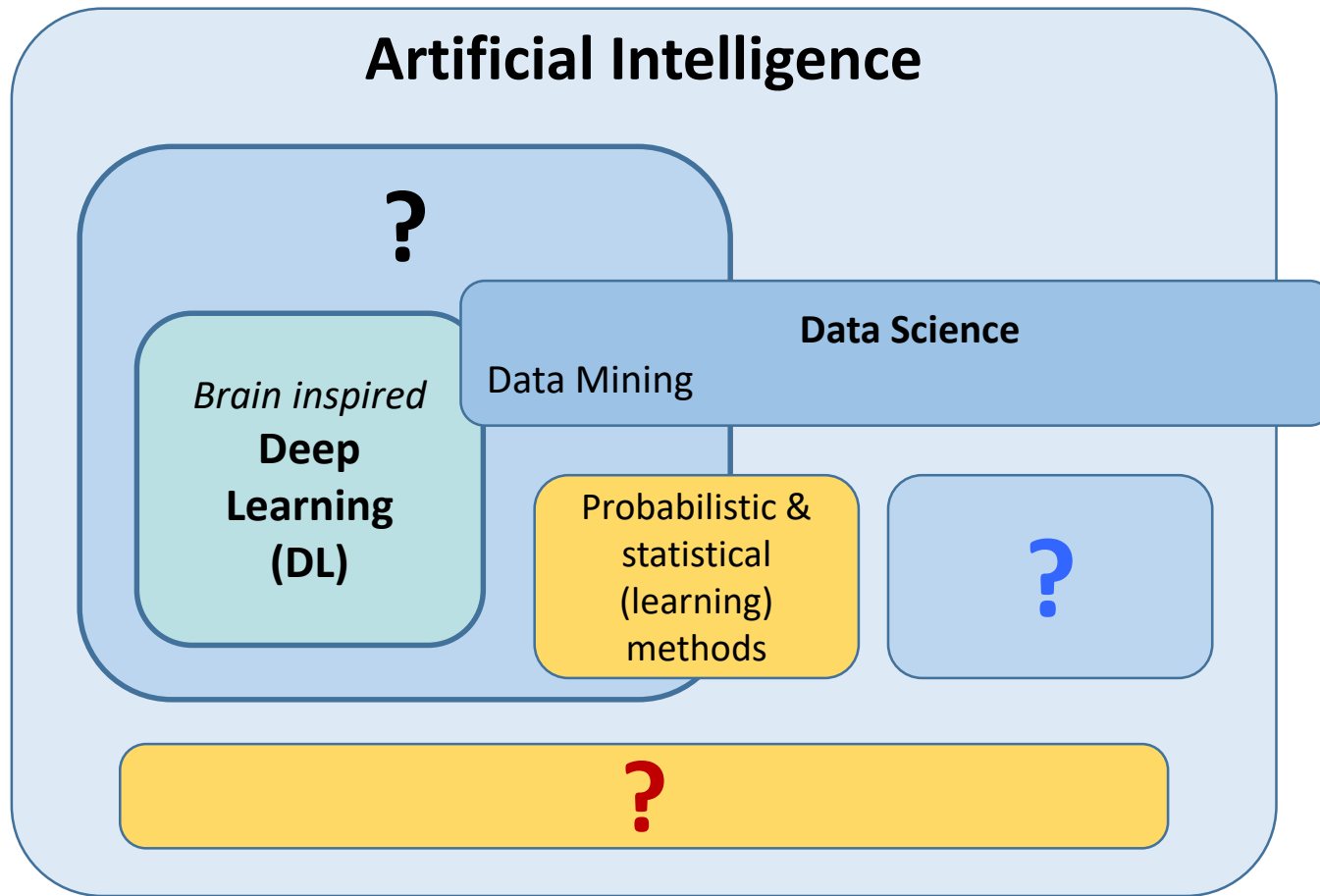
EDL Concept





REVIEW QUESTIONS

AI subfields



Genetic Algorithm (Evolutionary Algorithm)

initialize population

evaluate population

while TerminationCriteria_notSatisfied:

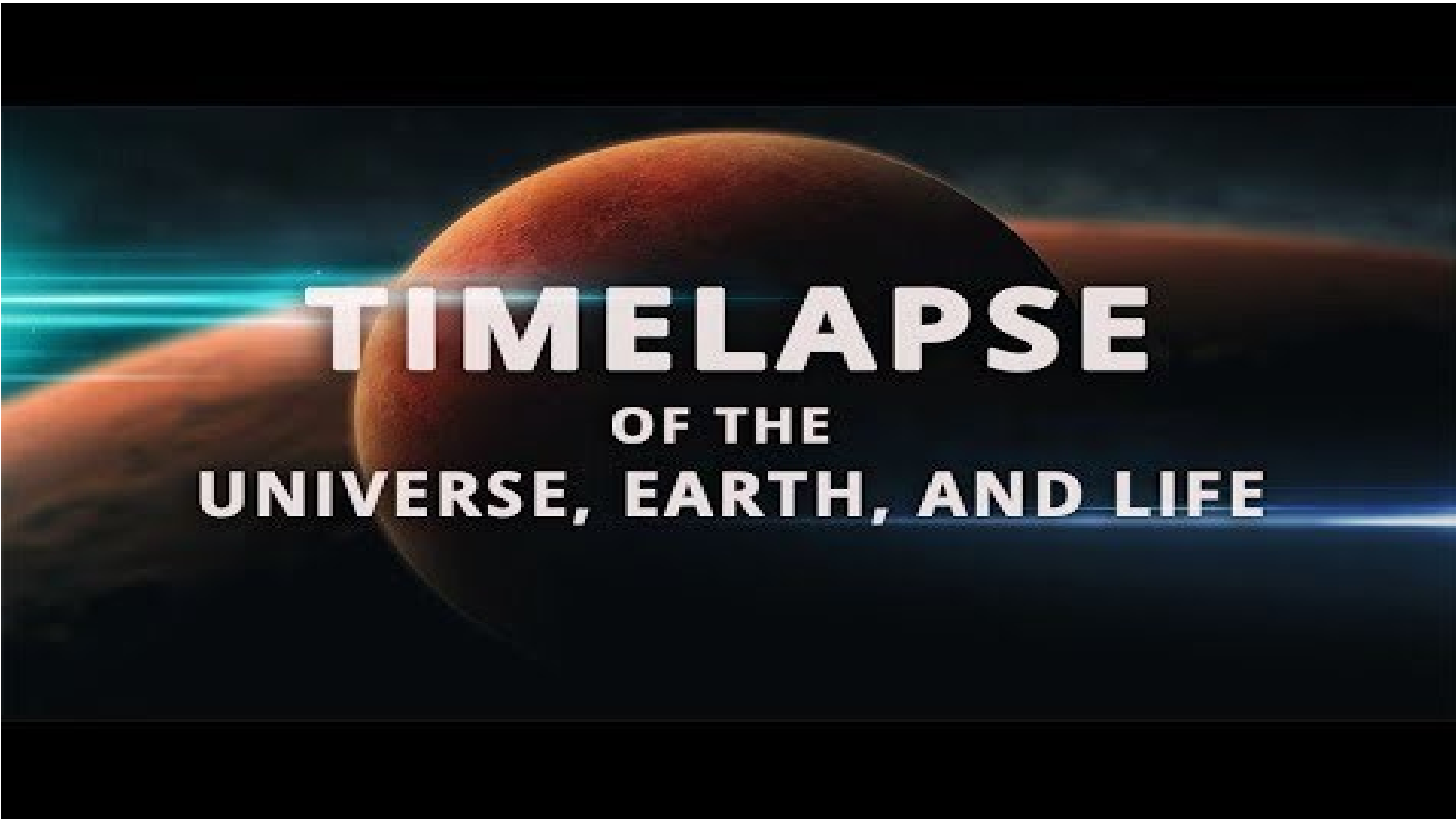


parents for reproduction

perform recombination (xover) and



ADDITIONAL VIDEOS TO WATCH



TIMELAPSE OF THE UNIVERSE, EARTH, AND LIFE

